SPM 5708 Homework 5 (Standardization, Normalization, & Paired Sample T-Tests)

Standardization means taking a value for a certain observation (e.g., a seasonal performance statistic for a certain player) and converting it into a ***z-score***. A z-score shows how many standard deviations above or below the population (group, league, etc.) average that specific value was. Z-scores are calculated as:

$$z = \frac{x - group\ mean}{group\ s.d.}$$

However, because z-scores aren't always the easiest thing to conceptualize or communicate, it can be helpful to take it one step further and ***normalize*** them so that they are presented on a scale that ranges from 0-100. Normally, we would have to do this by referencing our z-score in a z-table, but with the help of the *pnorm* function, we can normalize values with a few keystrokes. After normalizing a z-score, we see the percentage of observations that a specific value would be higher than in a normal group/population (i.e., the percentile it ranks at).

5.1 Comparing Mahomes to Hall of Famers

A lot of publicity surrounded Patrick Mahomes' second season in 2018 as he cruised to an NFL regular-season MVP title while leading a high-powered Chiefs' offense to the AFC Championship. Some have said his statistics were the best for a year-two player in NFL history, but making this comparison can be hard since the style of play in the NFL has dramatically shifted in recent years from a more run-oriented league to one dominated by the passing game.

Keeping in mind the shifts in play that have occurred over the decades, how do Mahomes' second-season stats compare to the second-season stats of NFL Hall of Famers Dan Marino and Kurt Warner? That is, which QB appeared to have the best second season, *given the era he played in*? By standardizing and normalizing the data found in the worksheets of the *Mahomes* file, we can compare these QBs across certain statistical categories.

1. The *Mahomes* Excel file contains three worksheets (*Mahomes, Warner,* and *Marino*) that you will need to import separately into R Studio. These sheets contain the passing statistics for every player who threw a pass in the 2018 (*Mahomes*), 1999 (*Warner*), and 1984 (*Marino*) seasons. We will need to subset the sheets appropriately before conducting our analyses. To do this, ***filter*** all three datasets so that they only contain players who are officially listed as quarterbacks (*Pos=="QB"*) ***and*** attempted at least 100 passes (*Att>=100*) in the observed season. This way, non-quarterbacks and backups are largely removed.

2. Standardize, and then normalize, the three players' values for completion percentage (*CmpPct*), adjusted net yards per attempt (*AdjNYPA*), and total passing touchdowns (*TD*) *relative to the QBs in their respective dataset*. To standardize, we run the raw values through the ***scale*** command. Once we get the standardized values (z-scores), we then use the *pnorm* function to get the normalized percentages (just like we did in class). **Report the normalized values for the three quarterbacks across these three statistics in a table below (you can create the table in Word if needed)**.

   ANSWER:

PLAYER	CmpPct Normalized	AdjNYPA Normalized	TD Normalized

Patrick Mahomes	0.6295893	0.9814836	0.9964723
Kurt Warner	0.9848153	0.9944321	0.9995395
Dan Marino	0.9409438	0.9964318	0.9998268

3. **Interpret (discuss)** your results as they relate to deciphering which QB (of the three) had the best second season, and which of the three statistical categories they were best in relative to each other and the QBs from their time period.

   ANSWER: We would argue that the best second season quarterback was Kurt Warner, he was clearly better than Mahomes and slightly better than Marino in completion percentage. And even though he was not better than Marino in the Adjusted Net Yards Per Attempt or Total Passing Touchdown classification, he was very close to him with a difference of 0.002 and 0.003 respectively. We could argue as well that Warner was better in these three categories in the year 1999 than at least 98.5% of the other QB's with more than a 100 attempts. Marino was better than 99.6% of the other QB's in the AdjNYPA and TD category, but better than 94.1% of the other players in CmpPct in the year 1984. And Mahomes, was only better than 63% of the other QB's in the year 2018 in the category CmpPct, while he was better than at least 98.1% of the other QB's in the 2 other categories.

5.2 Do Transfer Football Players Improve?

Paired-sample t-tests are hypothesis tests designed to compare whether *means* from two observations of the same entity (e.g., player or team) significantly differ. It is represented by the following hypothesis statements examining whether the mean difference is significantly different from zero:

H_0: $\mu_d = 0$

H_1: $\mu_d \neq 0$

Using the t-test, a ***p*-value** is calculated. If below our preferred alpha (α) of .05, we reject the null hypothesis (H_0) in favor of the alternative (H_1) hypothesis, which implies that the difference between the two means from the same subject is significantly different from 0. If the *p*-value is above .05, we fail to reject the null hypothesis and assume that the mean difference is not significantly different from 0.

1. Import the *fbs_portal.csv* file from Canvas that some of you used in *Homework 1*. Using R, apply a paired sample t-test to see whether the difference between transfer wide receivers' (*WR*) average pre-transfer (*PFF_Grade_Pre_WT*) and post-transfer (*PFF_Grade_Post_WT*) Pro Football Focus (PFF) grades is significantly different from 0 ($p < .05$). **Below, report and interpret your results in relation to your conclusions about this hypothesis test.**

   ANSWER: From the T-test below, we can see that the p-value is equal to 0.0399 which is less than the significant level 0.05, so we reject the null hypothesis. And we could say that there is a difference between PFF grades pre-transfer and PFF grades post transfer.

```
        Paired t-test

data:  WRs$PFF_Grade_Pre_WT and WRs$PFF_Grade_Post_WT
t = -2.059, df = 633, p-value = 0.0399
alternative hypothesis: true mean difference is not equal to
0
95 percent confidence interval:
 -1.24755167 -0.02956084
sample estimates:
mean difference
     -0.6385563
```

2. Now, conduct the same test for tight end transfers (*TE*). **Report and interpret your results below as they relate to your conclusions about this test.**

   ANSWER: As we can see from the output below, the p-value of 0.05785 is greater than our significant level 0.05, so we fail to reject the null hypothesis. This means that for tight end transfers, in average, the PFF grade pre transfer does not change much compared to the PFF grade post transfer. Nonetheless, the p-value is very close to our significance level and the mean difference is 1 unit, so it would be interesting to see the observations where position is TE more closely.

```
        Paired t-test

data:  TEs$PFF_Grade_Pre_WT and TEs$PFF_Grade_Post_WT
t = 1.9075, df = 205, p-value = 0.05785
alternative hypothesis: true mean difference is not equal to
0
95 percent confidence interval:
 -0.04141932  2.50623505
sample estimates:
mean difference
       1.232408
```

In the previous two examples, we were simply testing whether there was a significant difference in the average PFF grades of these FBS players before and after transferring. By not specifying a “direction” in our test (i.e., that the mean difference would be *positive* or *negative*), we were defaulting to what is called a ***two-tailed*** test. If our hypothesis is that PFF grades (performance) will significantly and positively *improve* following a transfer, we would need to utilize a ***one-tailed*** test to better account for the probability of detecting an effect in that direction. Note this initiates a slight shift in our underlying alternative hypothesis statement (H_1: $\mu_d > 0$).

3. For both the prior position groups (*WR* and *TE*), conduct one-tailed, paired sample t-tests that specifically examine whether the transfers' average PFF grades improve from pre- to post-transfer. This is where you will need to incorporate *alternative="less"* or *alternative="greater"* functions within the *t.test()* code, depending on whether you typed the pre-transfer (less) or post-transfer (greater) column first. **Report the results and your conclusions from both tests below.**

   ANSWER: Since our null hypothesis is H_0: $\mu_d <= 0$, and our alternative hypothesis is that H1: $\mu_d > 0$. Our first one-tail paired t-test was conducted for the WR group. As we can see from the output below, our p-value is 0.01995 which is less than the significance level 0.05. We reject the null hypothesis, and we can conclude that PFF grade increased on average post-transfer.

```
        Paired t-test

data:  WRs$PFF_Grade_Pre_WT and WRs$PFF_Grade_Post_WT
t = -2.059, df = 633, p-value = 0.01995
alternative hypothesis: true mean difference is less than 0
95 percent confidence interval:
       -Inf -0.1277008
sample estimates:
mean difference
     -0.6385563
```

   Since our null hypothesis is H_0: $\mu_d <= 0$, and our alternative hypothesis is that H1: $\mu_d > 0$. Our second one-tail paired t-test was conducted for the TE group. From the output we see a p-value equal to 0.9711 and so we fail to reject the null hypothesis because the p-value is greater than 0.05. Because we fail to reject the null hypothesis, this could mean that PFF grade for tight ends decreased on average post-transfer.

```
        Paired t-test

data:  TEs$PFF_Grade_Pre_WT and TEs$PFF_Grade_Post_WT
t = 1.9075, df = 205, p-value = 0.9711
alternative hypothesis: true mean difference is less than 0
95 percent confidence interval:
     -Inf 2.299951
sample estimates:
mean difference
       1.232408
```

4. Someone asks you to see whether FBS transfers who were 3-star recruits (*Star_Rating* column) average significantly **more** snaps per season after transferring. Using values in the *Snap_Count_Pre_per_Season* and *Snap_Count_Post_per_Season* columns, **provide an answer. Then, conduct the same test and provide the same answer for 5-star recruits.**

ANSWER: Our null hypothesis is H_0: μ_d <= 0, and our alternative hypothesis is that H1: μ_d > 0. We first did a one-tail t-test for the 3-star recruits. And from the output below we have that the p-value is greater than our significant level 0.05, so we fail to reject the null hypothesis. And from the mean difference it seems that 3-star recruits had less snaps after they transferred.

```
        Paired t-test

data:  star_3$Snap_Count_Pre_per_Season and star_3$Snap_Count
_Post_per_Season
t = 4.2672, df = 2969, p-value = 1
alternative hypothesis: true mean difference is less than 0
95 percent confidence interval:
     -Inf 27.80457
sample estimates:
mean difference
       20.06703
```

Our null hypothesis is H_0: μ_d <= 0, and our alternative hypothesis is that H1: μ_d > 0. Second, we did a one-tail t-test for the 5 star recruits. And from the output below we have that the p-value of 0.1392 is greater than our significance level 0.05 so we fail to reject the null hypothesis.

```
        Paired t-test

data:  star_5$Snap_Count_Pre_per_Season and star_5$Snap_Count
_Post_per_Season
t = -1.1081, df = 25, p-value = 0.1392
alternative hypothesis: true mean difference is less than 0
95 percent confidence interval:
     -Inf 28.99384
sample estimates:
mean difference
      -53.53846
```

SUBMIT THIS COMPLETED WORD DOC AND YOUR R SCRIPT TO THE ASSIGNMENT DROP BOX ON CANVAS WHEN FINISHED.